CANipulator

Dual CAN-Bus Interface

Documentation

v2

Introduction

The CANipulator is a versatile CAN bus interface designed for bridging, translating, baud-rate matching, and manipulating traffic between two CAN networks. Integrated CAN-FD transceivers support data rates of up to 5 Mbps, while jumper-selectable termination resistors and Molex DuraClik connectors simplify installation and integration.

By connecting two independent CAN networks or devices, the CANipulator enables a wide range of applications, including linking isolated networks, translating messages between peripherals and vehicle networks, integrating devices with otherwise incompatible systems, and performing baud-rate conversion for legacy hardware. Automotive-grade power regulation with reverse--polarity and transient protection ensures reliable operation in demanding electrical environments.

At the heart of the CANipulator is an Espressif ESP32-C5-WROOM-1U module featuring a 32-bit RISC-V processor running at up to 240 MHz. The module provides 384 KB of high-performance SRAM, 16 KB of low-power SRAM, 320 KB of ROM, 8 MB of Flash memory, and 8 MB of PSRAM. Built-in wireless connectivity includes dual-band Wi-Fi 6 (2.4 GHz and 5 GHz), Bluetooth LE 5, Zigbee, and Thread (IEEE 802.15.4).

An optional microSD card slot enables onboard data logging for long-term monitoring, diagnostics, and offline analysis.

Electrical Characteristics

Input Voltage: 5V to 36V (40V absolute max)

Typical Input Voltage: 10V to 15V

I/O Pin Voltage: 3.3V max

Operating current: ≈100mA (standard mode)

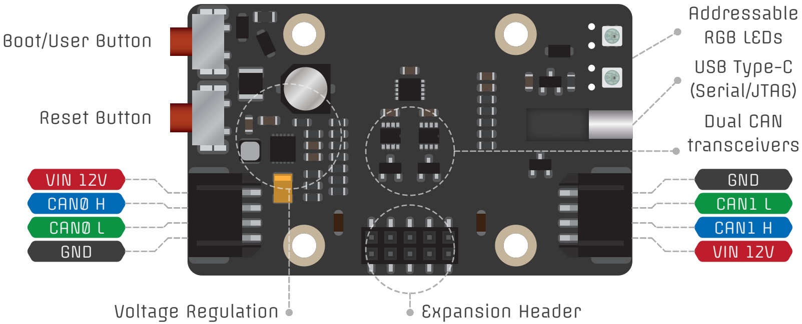
Operating temperature: -40°C to +85°C

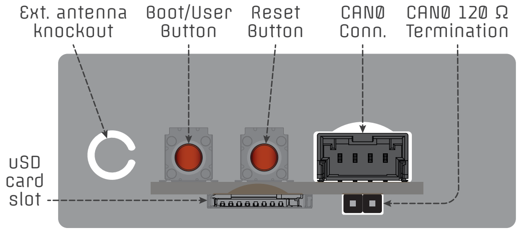
Maximum CAN transceiver speed: 5 Mbps

Notes:

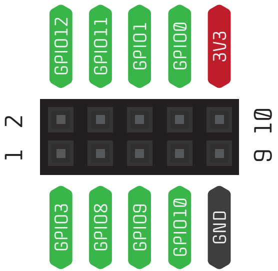
- Left/right VIN rails are connected
- CAN0/CAN1 Connector part numbers:
  Molex DuraClik pigtail 218323-1041; or
  Molex DuraClik female housing 560123-0400
  Molex DuraClik housing retainer 560125-0400
  Molex DuraClik crimp terminal 560124-0101
- Enclosure: Hammond Mfg. 1593KBK

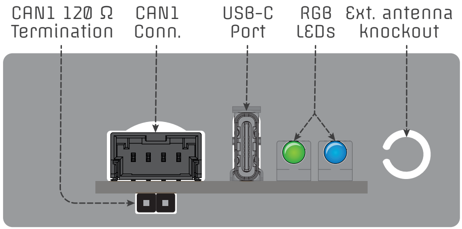
Overview





Expansion Header Pinout

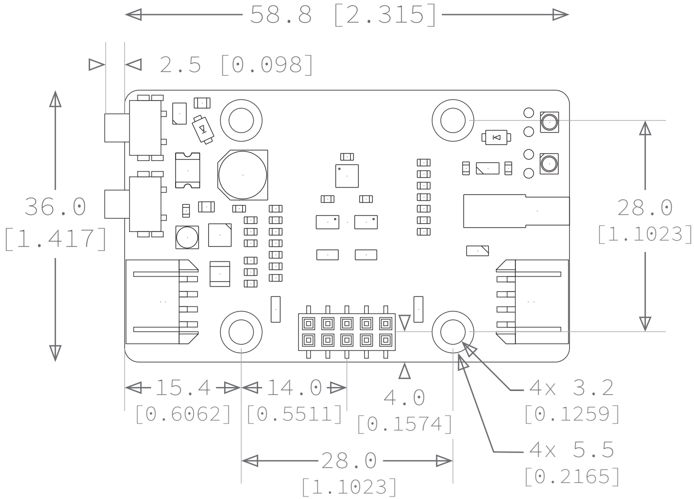




I/O Usage & Functions

GPIO0	I2C	I2C SDA - Connected to I/O Expander, with 4.7K pull-ip
GPIO1		I2C SCL - Connected to I/O Expander, with 4.7K pull-ip
GPIO2		Addressable RGB LEDs data pin
GPIO3		Unused - Connected to expansion header pin 1
GPIO4	CAN-BUS	CAN0 RX
GPIO5		CAN0 TX
GPIO6		CAN1 RX
GPIO7		CAN1 TX
GPIO8		Unused - Connected to expansion header pin 3
GPIO9		Unused - Connected to expansion header pin 5
GPIO10		Unused - Connected to expansion header pin 7
GPIO11		Unused - Connected to expansion header pin 4
GPIO12		Unused - Connected to expansion header pin 2
GPIO13	USB	USB D-
GPIO14		USB D+
GPIO23	microSD	uSD MOSI
GPIO24		uSD SCLK
GPIO25		uSD MISO
GPIO26		uSD Insertion Detect pin (low when inserted)
GPIO27		uSD CS
GPIO28		Boot / User button - Can be assigned function post-boot
EXP-IO0	CAN-BUS	CAN0 SIL - Drive high for silent mode, internal pull-down
EXP-IO1		CAN0 SHDN - Drive high for shutdown, internal pull-down
EXP-IO2		CAN1 SIL - Drive high for silent mode, internal pull-down
EXP-IO3		CAN1 SHDN - Drive high for shutdown, internal pull-down

Mechanical



Dimension units: mm [inches], Scale 1:1

Mounting hole pattern designed for: Hammond Mfg. enclosure 1593KBK

Recommended mounting screw size: M3×4mm or #4×1/4" self-tapping (1593ATS50)

Getting started: